

Marcus Forte - Full CV

A consolidated printable view with content from projects, education, skills, milestones, and contact.

About

Software Architect with an electrical engineering background and a passion for embedded systems, embedded Linux, mobile robotics, sensor fusion, and laser technology. Marcus develops high-quality, high-performance, and testable industrial software that integrates closely with hardware and equipment. As a former researcher in SLAM and autonomous vehicle localization, he brings a practical perspective on turning science into engineering products. He also creates [computer vision projects](#) and teaches software and technology through [Practical Software](#).

Projects

08/2024 - CURRENT

VANDERLANDE (VIA SIOUX TECHNOLOGIES)

SPO4x

Software Architect

Proposes and develops the multi-technology architecture for a high-speed sortation solution. Defines software interfaces, runs design reviews with R&D engineers, and advises embedded C and PLC programmers. A core embedded-Linux application communicates efficiently with more than 100 clients to control sort operations in real time; component emulation enables testing without a full physical system.

RESPONSABILITIES

- Propose and evolve a multi-technology architecture for high-speed sortation.
- Define software interfaces and drive design reviews with R&D engineers.
- Advise embedded C and PLC engineers on implementation trade-offs.
- Enable component emulation for testing without a full physical system.

C++14/C++17/C++20/C++23

C

Python

Data Definitions, gRPC, FlatBuffers

Networking

Embedded Linux

Software Architecture & Software Design

Low-Latency Communication

MEDIA

Video: <https://www.youtube.com/watch?v=uN-ywLXrYBc>

08/2024 - CURRENT

SIOUX TECHNOLOGIES

genAI Knowledge Management

Software Architect

Pilots and presents an internal enterprise system that assists with project and knowledge matching for customer problems, engaging company management and stakeholders.

LangChain

Python

Microsoft Azure

GitHub Copilot

Docker & Containers

LLM

React

08/2024 - CURRENT

SIOUX TECHNOLOGIES

AI-Assisted Development Training

Software Architect / Trainer

Delivers company-wide training on AI-assisted software development.

GitHub Copilot

AI-Assisted Development

Knowledge Sharing

04/2024 - 08/2024

SIOUX TECHNOLOGIES

Delta Embedded Linux Biosensing System

Senior Software Designer

Designed and developed an embedded Linux C++ application for photonics-based, label-less biosensing

Designed and developed an embedded Linux C++ application for photonics based, label-less biosensing equipment. The system controls infrared laser pulses and peripherals including temperature, pressure, and cylinder valves, while processing high-speed infrared-camera images. Built on a Yocto meta-project for the target image and supplied Python test scripts for test and integration engineers.

C++14/C++17/C++20/C++23

Data Definitions, gRPC, FlatBuffers

Yocto

C4 Architecture

OOAD

Instrumentation Engineering

Multi-Disciplinary Teamwork

MEDIA

Image 1: [/do.png](#)

Image 2: [/d1.png](#)

01/2023 - 04/2024

ASML (VIA ICT GROUP)

MATLAB Embedding Tooling

Software Designer

Developed and maintained tooling that generates a well-defined customized IDL interface for constrained customer environments, allowing MATLAB model embedding with verification, type checking, and local and remote testing. Later evolved the tightly coupled on-premises tool into a scalable, secure microservice with CI and DevOps practices.

MATLAB

C++14/C++17/C++20/C++23

Python

CI/CD

Domain-Driven Design

Jenkins

DevOps

03/2022 - 04/2024

ICT GROUP

Model-Based Engineering Expert Group

Software Designer

Evaluated Coco model-based development and Axini model-based testing with a digital twin of a miniature factory. Designed an architecture that enabled features to be modelled, tested in the digital twin, and deployed to physical equipment, with a focus on CI/CD integration.

C++14/C++17/C++20/C++23

C#

TypeScript / JavaScript

Jenkins

Data Definitions, gRPC, FlatBuffers

Digital Twins

01/2022 - 01/2023

ASML (VIA ICT GROUP)

ASML (VIA ICI GROUP)

MATLAB Embedding Tooling

Software Engineer

Developed and maintained tooling that generates a well-defined customized IDL interface for constrained customer environments, allowing MATLAB model embedding with verification, type checking, and local and remote testing. Later evolved the tightly coupled on-premises tool into a scalable, secure microservice with CI and DevOps practices.

MATLAB

C++14/C++17/C++20/C++23

Python

CI/CD

Domain-Driven Design

Jenkins

DevOps

10/2019 - 04/2024

DUNA SYSTEM

Nomad Portable LiDAR Scanner

CTO and Co-Founder

Designed hardware and software for a portable LiDAR-based 3D scanning device, including a custom SLAM architecture. Addressed sensor drivers, data messaging and retention, real-time processing, and operator visualization under battery-life and embedded-computing constraints.

C

C++14/C++17/C++20/C++23

ROS and ROS2

Python

Sensor Fusion

MEMS and LiDAR Sensors

Point Cloud Processing

Docker & Containers

MEDIA

Video: <https://youtu.be/EdXV7VaUuRo?si=AwFDemyK5Fi41vSa&t=59>

10/2019 - 04/2024

DUNA SYSTEM

Point-Cloud Web Platform

CTO and Co-Founder

Architected and partly implemented a scalable web platform for streaming, visualizing, processing, and exporting point clouds, including source data over 10 GB. Led a multidisciplinary team of more than seven colleagues and shared progress with investors.

C

C++14/C++17/C++20/C++23

TypeScript / JavaScript

Databases: MongoDB, SQL

CI/CD

Technical Team Leadership

Point Cloud Processing

09/2021 - 12/2021

INSTITUTO ATLÂNTICO

Cyber-Physical Systems and IoT Solutions

Cyber Physical Systems Specialist and Senior Software Developer

Advised innovation business goals for cyber-physical products. Developed sensor-data logging and transmission algorithms for a LoRa-connected cycling IoT device, and contributed to a lightweight secure remote-desktop application for low-end laptops connected to high-performance computing.

C++14/C++17/C++20/C++23

Sensor Fusion

IoT

TypeScript / JavaScript

SOLID Architecture

Technical Team Leadership

11/2020 - 08/2021

SENAI

Industrial Automation Consultancy

Consultant of Technical and Technological Services

Developed industrial solutions related to sensory instrumentation, control, and automation.

Control Systems

Technical Team Leadership

10/2018 - 10/2020

GPAR (PARTNERSHIP WITH EDP)

Autonomous Coal Stockpile Monitoring

R&D Engineer

Developed a solution for estimating coal-stockpile inventory using an autonomous UAV with laser scanning and GPS/RTK georeferencing. The workflow automated flight, 3D capture, landing, point-cloud processing, visualization, and selected-region volume estimation.

C

C++14/C++17/C++20/C++23

Python

ROS and ROS2

UAV

MEMS and LiDAR Sensors

Sensor Fusion

MEDIA

Video: <https://youtu.be/yfJFPgxtBzU?si=Pf5jlf2HLnPm-wG-&t=89>

07/2016 - 08/2017

07/2016 - 02/2017

ENCCEL AUTOMÁTICO

Automatic-Door Control Prototyping

Hardware and Software Prototyping Engineering Intern

Designed hardware and software for an electronic automatic-door control system.

C

C++14/C++17/C++20/C++23

Personal Projects

Portable 3D Scanner

Developed a portable 3D scanner for capturing real-world objects and creating 3D models in real time. The scanner uses a custom-made, decoupled SLAM algorithm that turns a sensor fusion method into an elegant non-linear optimization problem. The scanner device is as minimal as possible, with a single 3D LiDAR sensor with integrated IMU, a single-board computer, and a battery.

C

C++14/C++17/C++20/C++23

Python

Data Definitions, gRPC, FlatBuffers

Sensor Fusion

MEMS and LiDAR Sensors

Point Cloud Processing

MEDIA

Video: https://youtu.be/xEbS_dFyPWo

Practical Software

Educational content initiative focused on practical software-engineering topics, architecture, and AI-assisted development workflows.

The idea came out of a passion to teach and share knowledge about software engineering, embedded systems, and robotics, that started in the very early days of my career as an assistant mentor for the course of embedded system back in university.

The youtube channel sediments one of my life's drives: *'Learn → Apply → Teach.'*

RESPONSABILITIES

- Design and publish practical content on software architecture and engineering workflows.
- Teach embedded systems, robotics, and AI-assisted development practices.
- Translate complex technical topics into actionable lessons for developers.

Clear Communication

Knowledge Sharing

C++14/C++17/C++20/C++23

Python

MEDIA

Video: <https://www.youtube.com/watch?v=-kvrorbjzco>

Education

08/2017 - 08/2018

Master's degree

Federal University of Ceará

Electrical Engineering, focused on Automation, Control and Robotics. Master thesis: Reference trajectory tracking control of a nonholonomic mobile robot with inertial sensor fusion.

08/2014 - 08/2015

Master's Program

Technische Universität Dresden

Nanoelectronics Systems, exchange program.

02/2011 - 07/2017

Bachelor's degree

Federal University of Ceará

Electrical Engineering.

Skills

Design

Software Architecture & Software Design (8/10)

Data Definitions, gRPC, FlatBuffers (10/10)

C4 Architecture (9/10)

Domain-Driven Design (7/10)

SOLID Architecture (7/10)

Technology

C++14/C++17/C++20/C++23 (10/10)

Embedded C++ (8/10)

C (8/10)

Python (9/10)

TypeScript / JavaScript (7/10)

MATLAB (7/10)

Linux (9/10)

CUDA / OpenCL (7/10)

Docker & Containers (10/10)

CI/CD (8/10)

Git / GitHub (10/10)

Databases: MongoDB, SQL (7/10)

LaTeX (8/10)

Networking (6/10)

LangChain (8/10)

Database: MongoDB, SQL (7/10)

Python (8/10)

Networking (8/10)

Programming (8/10)

Microsoft Azure (5/10)

GitHub Copilot (9/10)

LLM (9/10)

Yocto (2/10)

Jenkins (8/10)

C# (5/10)

UAV (7/10)

People

Knowledge Sharing (10/10)

Clear Communication (10/10)

Multi-Disciplinary Teamwork (10/10)

Scrum & Agile (8/10)

Technical Team Leadership (9/10)

Innovation & Tech Business (9/10)

Technique

Computer Vision (9/10)

Point Cloud Processing (10/10)

ROS and ROS2 (10/10)

MEMS and LiDAR Sensors (10/10)

Embedded Linux (10/10)

RTOS (7/10)

SLAM (Simultaneous Localization and Mapping) (8/10)

Sensor Fusion (8/10)

Mobile Robotics (8/10)

IoT (8/10)

Low-Latency Communication (7/10)

AI-Assisted Development (10/10)

OOAD (9/10)

Instrumentation Engineering (5/10)

React (4/10)

DevOps (9/10)

Digital Twins (6/10)

Control Systems (9/10)

Languages

Portuguese: 10/10

English: 9/10

German: 3/10

Dutch: 1/10

Qualifications & Trainings

- 2023 - Technical Architect, ICT Group
- 2023 - Architect in The Organization, ICT Group
- 2022 - Software Domain-Driven Design, ICT Group
- 2022 - Thinking Critically and Talking Clearly, ICT Group
- 2017 - Solar Panel Project Design Course, FOTAIC

Additional Milestones

- 2024 - Started Practical Software, a YouTube channel for sharing software-engineering knowledge.
- 2024 - Started Algosphere, a meetup group for algorithm developers.
- 2024 - Recorded the Unlocking Sensor Fusion podcast episode with Inspiring Computing.
- 2022 - Proposed and initiated an EUR 80,000+ innovation project: Fusing AI and Computer Vision for Classification of BIM entities in 3D Point Clouds.
- 2021 - Proposed Portable & Scalable 3D Scanning Solution for Instituto Atlântico's PRAIA acceleration program.
- 2020 - Proposed and initiated the EUR 15,000+ Industry 3D Scanning innovation project.
- 2018 to 2020 - Technological Consultant of the Multidisciplinary Extension Program (RAITEC).
- 2017 to 2019 - Conducted an ARM Cortex learning course during SEEL Tech Week.

Contact

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YouTube: [youtube.com/@practicalsoftwaremarcus](https://www.youtube.com/@practicalsoftwaremarcus)
